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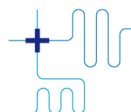
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Fruit Yield Estimation and Forecasting for Precision Agriculture

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Abstract. Precision agriculture is an ever-growing domain. Fruit yield estimation and forecasting is a necessary step towards precision agriculture. Reliable and accurate estimation of fruit yield in an orchard help the farmers to make suitable arrangements. Fruit yield estimation is performed on video footage in which fruits are detected and tracked in each frame of the input video. YOLO V5 (You Only Look Once Version 5) neural network are applied to detect fruits in an orchard. Bounding boxes are extracted from detection and Non-Maximum Suppression (NMS) is performed to avoid multiple detections. Then the results are input into the tracking pipeline. DeepSORT algorithm is used for tracking the fruits. By keeping track of the fruits throughout the video frames, it is ensured that we are estimating the total fruit count appropriately when they are detected. Future fruit yield prediction is also carried out using time series forecasting by ARIMA model. This work may greatly help the needy farmers in yield estimation and predicting the future yields for sustainable growth.

INTRODUCTION

In most countries, fruit cultivation is practiced in large areas to meet the worldwide demand. Because manual counting involves skilled observation and longer estimating times, it has low accuracy results and is expensive. With manual yield estimation, subsequent decision-making for farmers becomes a difficult task. If farmers know their yield, they'll accurately calculate their harvest [1] and production costs, and their profits, and founded their equipment prior to. These problems drive the development of an intelligent fruit yield estimation system that gives farmers greater advantages when choosing when to harvest, market their produce, etc. If farmers know their yield, they will accurately calculate [1] their harvest and production costs, and their profits, and founded their equipment ahead. Any farmer is inquisitive about knowing what quantity yield he's getting ready to expect. Knowing the yield help farmers make informed choices in terms of the way to allocate resources for harvesting, preparing storage per fruit sizes, and better plan for harvest routes and packaging and to give some thought to the methods to extend their yield. This could benefit farmer's profitability and efficiency [1].

RELATED WORKS

This section describes previous research and systems on fruit yield estimation. Fruit recognition could be a fundamental step in autonomous yield calculation. The objective is to identify every fruit in an image. A novel Faster R-CNN model with classifier fusion to automatically detect small fruits [2]. To learn three classifiers for object categorization, it uses features from three separate layers. A straightforward convolutional layer combines classifier probabilities. The region proposal network is trained using a unique loss function with classifier correlation.

Multi-task cascaded convolutional networks-based detector containing three cascaded networks [3]. Fusion augmentation, an improved augmented method, a procedure that is based on image fusion. The average time cost of the detector was less than 80 seconds per one hundred images [3-5].

A novel weakly-supervised framework for fruit counting. The WSCOUNT strategy in which novel convolutional neural [4] network architecture that requires only a image level binary classifier. Exploits a binary presence absence classifier (PAC), trained with only image level labels [4].

YOLO V3 is employed with spatial pyramid pooling (YOLOv3-SPP) as the back-end object detector [6-9]. The CNN model used by YOLO V3 is called DarkNet53 and it uses 53 layers. The anchor box that intersects the real world most closely. The ideal forecast is the bounding box. After all boxes have been checked, non-max suppression (NMS) is predicted. An improved YOLO-V4 deep learning algorithm is used for detection [5]. The CNN model used by YOLO V3 is called DarkNet53 and it uses 53 layers [10]. The anchor box that intersects the real world most closely. The ideal forecast is the bounding box. After all boxes have been checked, non-max suppression (NMS) is predicted.

Deep learning architectures have recently been utilised to handle a number of computer vision issues, including gesture recognition, object detection, image segmentation, and action recognition. When compared to traditional methods based on hand-crafted features, deep learning techniques do particularly well when processing video streams to distinguish individual activities. The secret to these results is a rich and discriminative representation extracted from multi-layer nonlinear systems in the deep learning approaches, despite the fact that numerous ways to extract superior handmade features have been offered. Convolutional neural networks (CNN) and multi-layer fully connected neural networks (deep neural networks) were used by us to find significant time-space features, and prior works demonstrated the potential of the deep learning method for estimating fruit output [11-14].

Deep learning has brought attention thanks to its ability to supply higher accurate models for object detection, object classification, scenario analysis, real-time surveillance, ADAS, etc. Models in deep learning are trained with high-quality images to predict and provide final high-quality output. So, lesser quality image causes the computer vision algorithms to affect their performance, providing undesirable output. With high-quality images, the DL algorithms learns the features that are necessary for performing the applications efficiently. Therefore, deep learning techniques using YOLO V5 are chosen for fruit yield estimation.

METHODOLOGY

The proposed system is fruit yield estimation and forecasting for precision agriculture. Fruit yield estimation is done with use of YOLO V5 and DeepSORT. Future fruit yield prediction is carried out using time series forecasting by ARIMA model. This work may greatly help the needy farmers in yield estimation and predicting the future yields for sustainable growth. The project is done on mango fruits.

Fruit Yield Estimation

Fruit yield estimation is performed on video footage in which fruits are detected and tracked in each frame of the input video. Detection is performed with the use of YOLO V5 (You Only Look Once Version 5) neural network. By keeping track of the fruits throughout the video frames [1], it is ensured that we are estimating the total fruit count appropriately when they are detected. The final count generated gives the estimate of the total number of fruits present in the video.

Dataset Collection

The dataset containing images of mangoes is collected. The Mango dataset comprises jpeg images showing mango fruits in trees. This type of dataset is required for fruit yield estimation since the dataset contains mango fruits in highly occluded conditions. This will increase the accuracy of the model to detect in highly complex environments.

Dataset Labeling

Labeling of dataset is done using labeling tools. There's a limited amount of annotated data currently available for object detection. That can be a hurdle. If we are not able to find an annotated dataset then we need to create one by ourself. There are many tools available on GitHub that we can use to annotate the images for free.

Dataset are labeled using Labellmg tool. Labellmg supports YOLO and PASCAL VOC formats. Since the project is predicated on YOLO V5. The format employed in the project is YOLO format. Annotations are saved as text files in YOLO format.

In this case, since mango dataset is collected, the fruits are labeled as 'mango' as shown in Figure 2. The mango fruit images are annotated as "mango" by drawing bounding box around it. Hence just one class i.e "mango" class is present. therefore, the mango class has class ID as 0. The data set contains mango fruits which are highly occluded by leaves, overlapped mangoes, mangoes at different illumination conditions. Hence through labeling mangoes in such complex environments increases the detection mechanism of the model.

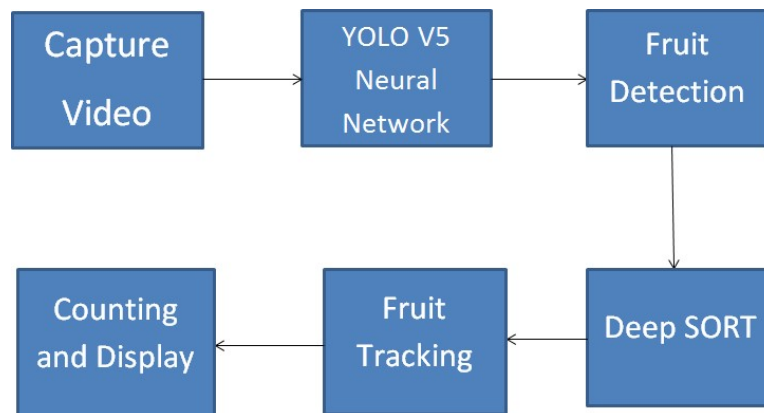


FIGURE 1. Fruit Yield Estimation

Lighting and occlusion [1] both pose difficulties in detection and tracking. The dataset is manually annotated with class name as "mango". So, in this project, only one class is required and used i.e 'mango' class with class numbered as '0' since the number of classes is '1'.

Building YOLO V5 Model

The YOLO V5 model is a object detection network. The purpose the YOLO V5 model is for detecting mangoes. The YOLOv5 model consists of mainly three parts: backbone network, neck network, and head layer. YOLO V5 utilizes PyTorch. The backbone layer is used for feature extraction. CSPDarknet53 is used as the backbone of YOLO V5. The backbone layer also consists of focus layer which is used for performing slicing operations of the input image. The neck layer is used for generating feature pyramids. The mangoes with different scales can be identified with feature pyramids. The head (YOLO) layer is the output detection layer from which the final detections of the input image are produced with bounding boxes. YOLO V5 perform Non-Maximum Suppression (NMS) to avoid multiple detections to get the final mango detections.

After labeling of dataset is done, the YOLO V5 model is trained with the custom dataset i.e Mango dataset. Split the dataset into training and testing dataset by 70% and 30% respectively. Training of YOLO V5 is done in Google Colab since GPU is required during training. Training Train custom mango dataset with YOLOv5 in Google Colab. YOLOv5s is used due to its smallest and fastest features. Clone the YOLO V5 from github repository. Then install required requirements. YOLO V5 contains essential inbuilt functions required for effective running of the project.

In YOLO v5, coco128 dataset containing 80 classes are pretrained with YOLO V5. For these classes, the pretrained weights are used for detection in such cases. But mango is not pretrained with YOLO. So, the mango dataset collected have to be trained with YOLO V5 to generate the weight file used for mango detection. YOLO V5 is fine-tuned by training with dataset comprising of 101 mango tree images. Train YOLOv5s model on COCO128.yaml for 100 epochs by specifying custom dataset path, batch-size as 16, image size as 224, and with pretrained –weights yolov5s.pt. Pretrained weights are auto-downloaded from the latest YOLOv5 release.

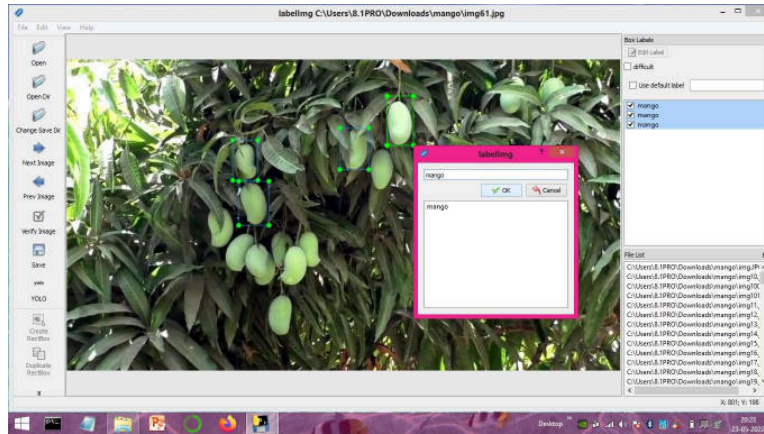


FIGURE 2. Labeling using LabelImg

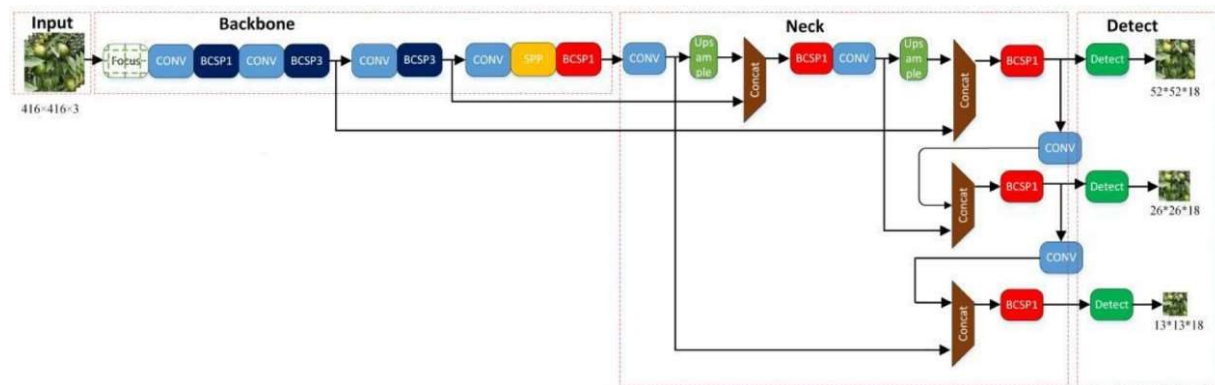


FIGURE 3. Architecture Of YOLO V5

The model is trained by 100 epochs with batch size 16. It took 0.041 hours for completing 100 epochs. The model summary obtained represents the features of the model comprising its layers, parameters, gradients, etc. The Model Summary obtained is: 213 layers, 7012822 parameters, 0 gradients, 15.8 GFLOPs.

Testing After YOLO V5 model training is done, the model is tested. Testing is done in Jupyter Notebook since the results are stored from the drive to the system. The YOLO V5 custom trained model is tested against the images containing mangoes in the dataset.

The results of testing show the number of images tested and its dimensions. It also shows the count of the number of mangoes detected. It also outputs the speed required for the detection which is just milliseconds taken for detecting and the inference and Non-Maximum Suppression (NMS) per image at shape of the image. The result shows the mango fruits detected by bounding box and the confidence score of each detected mango fruit is also displayed.

Tracking Using DeepSORT

DeepSORT is used for fruit tracking after detections of fruits are done in the input video. It also uses the Kalman filter to predict the current position using the last location of the fruit. [6] The highest identifier obtained during tracking is used to estimate the total count of the mangoes in the input video. Then DeepSORT associates between the [1] same fruits in different frames by using distance metrics, based on the motion of the fruit and on the appearance of the fruit. After we obtain the first bounding boxes from YOLO [1], the first step in the DeepSORT pipeline is initializing the tracked object for the detections. The track is given a number of attributes including Track ID as an identifier, Feature matrix which is the output of the OSNet and represents the object is appearance, Track state which represents the current state of the track, mean and covariance matrices which are used for the Kalman Filter that are updated with every prediction. [1].

The YOLOv5-DeepSort-OSNet model is used for fruit tracking in this project. Track Identifier (ID) is mainly used for estimating the fruit yield. It contains many inbuilt files which are beneficial for tracking.

Fruit Counting for Yield Estimation

Fruit yield estimation is done from the results obtained after tracking. The results of tracking are stored to the google drive containing the weights folder of exp2 in the YOLO v5 folder. Out.txt generated as a result of tracking are used for fruit yield estimation. The out.txt contains frame number of the in out video, tracking unique identifier (ID), bounding box coordinates and the class of the detected object. Here only one class is generated ie class '0' since only mango is labeled in the custom dataset. Out of the columns generated, drop all other columns except second columns which represents the tracking identifier to get the fruit count. From the second column (i.e. Track identifier), the highest value of the tracking identifier represents the count of the mangoes in the input video.

Fruit Yield Forecasting

Fruit Yield Forecasting is carried out after fruit yield estimation by fruit counting is done to predict the future production trend. This work may greatly help the needy farmers in yield estimation and predicting the future yields for developing a sustainable future. Fruit Yield Forecasting is done with ARIMA model. Auto Regressive Integrated Moving Average (ARIMA) is an algorithm for forecasting that is based on the previous values of the dataset in order to predict the future values. ARIMA is an efficient technique for time series forecasting.

Dataset Collection and Preprocessing

In this project, dataset for future forecasting comprising Indian status of mangoes is collected from Indiatat as shown in Figure. All India Area, Production and Productivity of mango in India of years 1987-1998, 1991-1992 to 2021-2022 are collected from Indiatat. Data Cleaning is done in order to handle missing data. Drop observations that have missing and null values. Columns containing percentage of fruit area and production having majority of null values are eliminated.

Stationarity Checking

The primary step of the ARIMA model is to make the dataset stationary in order to build an ARIMA model. The stationarity of data is tested by Augmented Dickey Fuller (ADF) test using test-stationarity function. Test Statistic value is greater than critical value. So, dataset is not stationary. To make data stationary, most common technique is used that is to subtract the past production (in MT) value from the present production (in MT) value in the dataset.

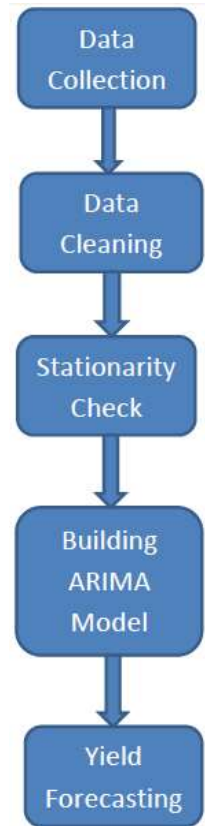


FIGURE 4. Fruit Yield Forecasting

Then test for stationarity of the data again after performing differencing. After first order differencing(d), Test statistic becomes less than critical value as shown in figure. So, data becomes stationary.

Building ARIMA Model

After time series dataset are made stationary, ARIMA model is builded. With p and q standing for the order of the Auto-Regressive (AR) term and Moving Average (MA) term, respectively, Auto ARIMA is used to determine the order of ARIMA(p,d,q) to be utilised to forecast the best results. The time series [7] must have d differences in order for it to remain stationary.

The optimal parameters, p, q, and d, are chosen via auto ARIMA. We utilise the fit() command in Auto ARIMA to fit the model without having to choose the p, q, and d combination. The optimal p, q, and d parameter combinations are determined by the Auto Arima. In order to choose the ideal set of parameters, Auto ARIMA also looks at the AIC and BIC values [8] that are created. Values of the Akaike Information Criterion (AIC) [9], Bayesian Information Criterion (BIC), Hannan Quinn Information Criterion (HIC) are used to compare ARIMA models. The model is better when these values are lower.

Forecasting Fruit Yields

From the obtained Seasonal ARIMA (p,d,q)x(P,D,Q)_m is used to forecast the future yields. A graph is plotted between actual production and predicted production to determine how much the model fits the data. As the ARIMA model built fits data approximately, the model used to forecast the yield for the next five years ie from 2022-2026. The graph is plotted and the predicted production (in MT) are displayed showing the yield forecasted.

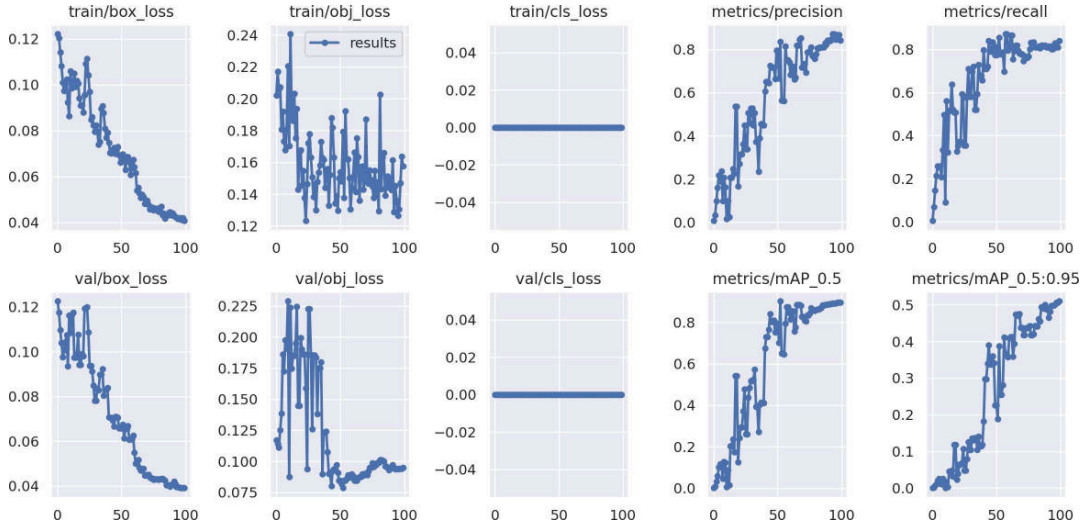


FIGURE 5. Training results showing Precision, Recall, mAP curve, etc.

EVALUATION CRITERIA AND RESULTS

The evaluation metrics used in the fruit yield estimation system is evaluated by using Precision, Accuracy, F1 score and mean Average Precision (mAP).

$$Precision = (TP) / (TP + FP) \quad (1)$$

$$Recall = (TP) / (TP + FN) \quad (2)$$

$$F1 \text{ measure} = (2 * Precision * Recall) / (Precision + Recall) \quad (3)$$

where T P is True positive sample [10] and FP is False positive, denote positive and negative sample respectively, predicted to be true and FN is False negative.

Precision of 82.4, Recall of 87.2, F1-score of 71.9, mAP of 89.6 is obtained with the fine-tuned weights of YOLOV5 trained with our custom prepared mango dataset. Training results are plotted as results.png after training completes as shown in Figure 5.

The tracking results are stored in the weight file present in the runs folder of YOLO. The track folder saves the tracked video and tracking information as out.txt. The out.txt contains information such as frame number, unique identifier, bounding box coordinates and the class number of the tracked object ie. mango.

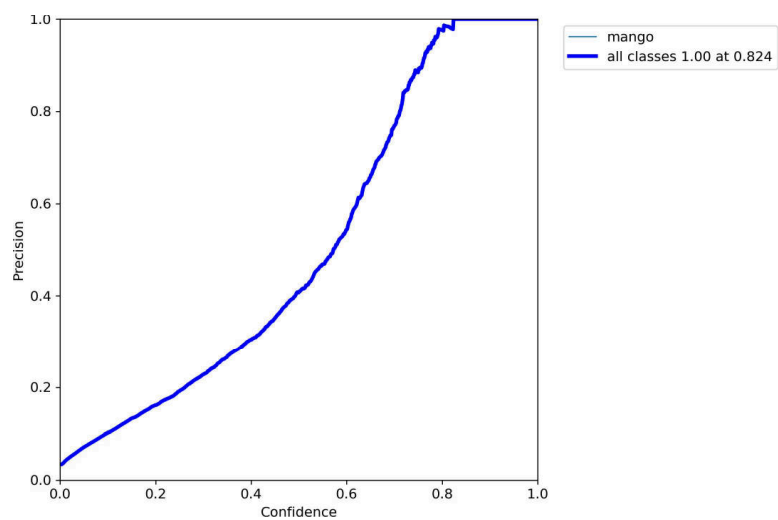


FIGURE 6. P curve

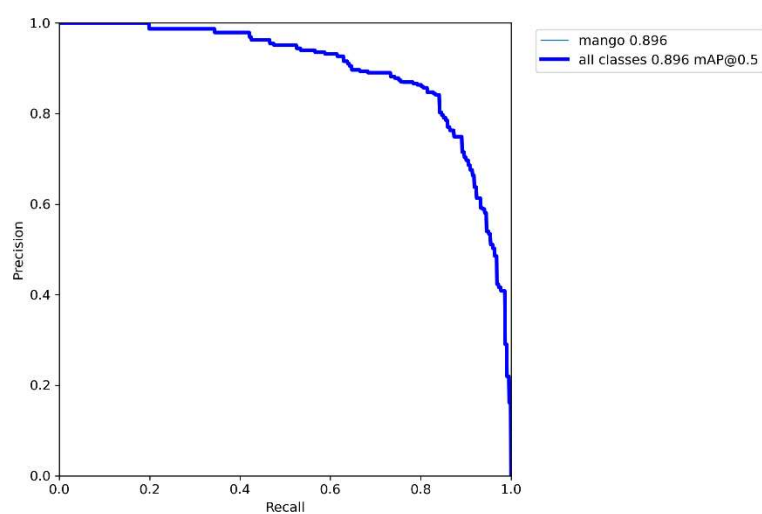


FIGURE 7. PR curve

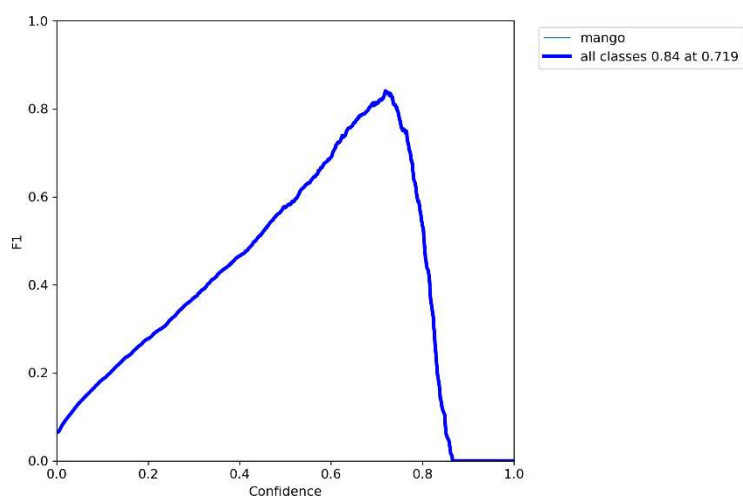


FIGURE 8. F1 curve



FIGURE 9. Fruits detected using YOLO V5

```
image 1/1: 720x1280 5 mangos
Speed: 379.4ms pre-process, 1965.4ms inference, 293.3ms NMS per image at shape (1, 3, 384, 640)
```

FIGURE 10. Detection Results

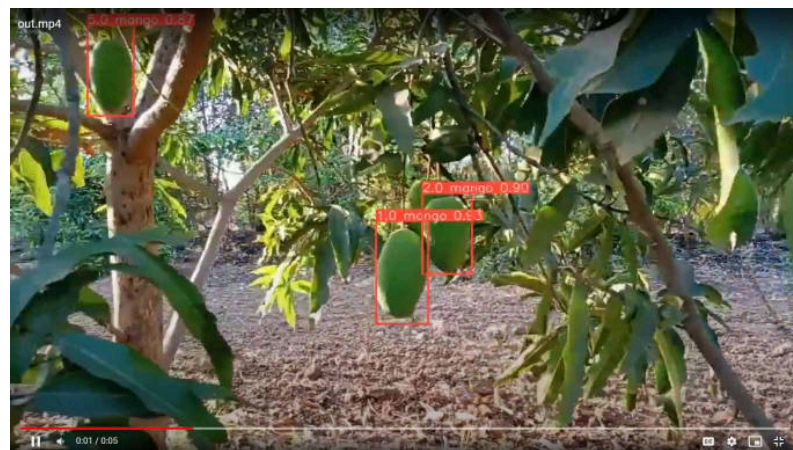


FIGURE 11. Tracking of mango in the input video

```

106 2 929 301 87 136 -1 -1 -1 0
106 3 411 73 66 122 -1 -1 -1 0
106 5 0 81 100 191 -1 -1 -1 0
107 1 846 346 96 167 -1 -1 -1 0
107 2 928 305 91 129 -1 -1 -1 0
107 3 387 61 65 123 -1 -1 -1 0
107 5 0 84 105 185 -1 -1 -1 0
108 1 846 346 98 168 -1 -1 -1 0
108 2 932 305 88 128 -1 -1 -1 0
108 3 396 65 71 126 -1 -1 -1 0
108 5 0 85 105 182 -1 -1 -1 0
109 1 851 345 96 167 -1 -1 -1 0
109 2 937 305 84 127 -1 -1 -1 0
109 3 377 55 71 128 -1 -1 -1 0
109 5 0 85 107 181 -1 -1 -1 0
110 1 858 347 98 163 -1 -1 -1 0
110 2 944 303 86 130 -1 -1 -1 0
110 3 406 66 65 122 -1 -1 -1 0
110 5 3 90 116 178 -1 -1 -1 0
110 8 0 402 91 141 -1 -1 -1 0
111 1 861 354 102 155 -1 -1 -1 0
111 2 947 306 90 130 -1 -1 -1 0
111 3 398 61 64 123 -1 -1 -1 0
111 5 11 95 114 178 -1 -1 -1 0
112 1 865 356 100 152 -1 -1 -1 0
112 2 952 307 87 131 -1 -1 -1 0
112 5 17 98 112 174 -1 -1 -1 0
112 8 0 387 107 155 -1 -1 -1 0
113 1 869 349 93 162 -1 -1 -1 0
113 2 955 303 85 137 -1 -1 -1 0
113 3 419 64 62 130 -1 -1 -1 0

```

FIGURE 12. Results of Tracking

```
print("total number of mangoes : {}".format(max_value))
```

```
total number of mangoes : 8
```

FIGURE 13. Results of counting (yield estimation)

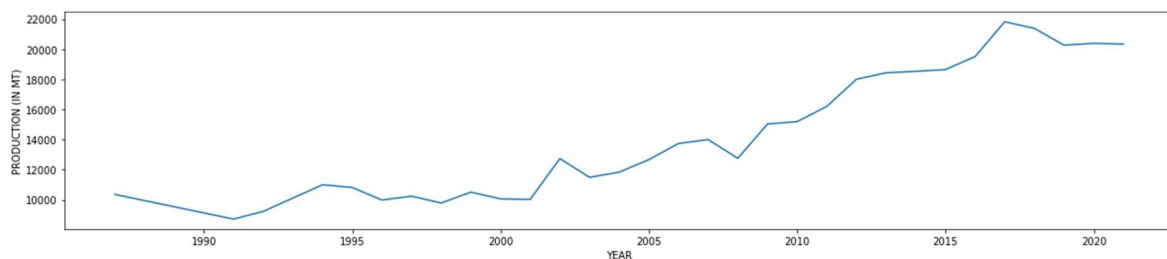


FIGURE 14. Plotting production (in MT) per year

SARIMAX Results						
Dep. Variable:	PRODUCTION (IN MT)			No. Observations:	32	
Model:	ARIMA(0, 1, 1)x(0, 0, 1, 12)			Log Likelihood	-255.919	
Date:	Tue, 05 Jul 2022			AIC	517.839	
Time:	15:37:19			BIC	522.141	
Sample:	0			HQIC	519.241	
	- 32					
Covariance Type:	opg					
	coef	std err	z	P> z	[0.025	0.975]
ma.L1	-0.1278	0.152	-0.839	0.401	-0.426	0.171
ma.S.L12	0.4476	0.285	1.570	0.116	-0.111	1.007
sigma2	8.519e+05	2.28e+05	3.731	0.000	4.04e+05	1.3e+06
Ljung-Box (L1) (Q):			0.04	Jarque-Bera (JB):	0.77	
Prob(Q):			0.84	Prob(JB):	0.68	
Heteroskedasticity (H):			3.57	Skew:	0.39	
Prob(H) (two-sided):			0.06	Kurtosis:	3.00	

FIGURE 15. ARIMA modeling

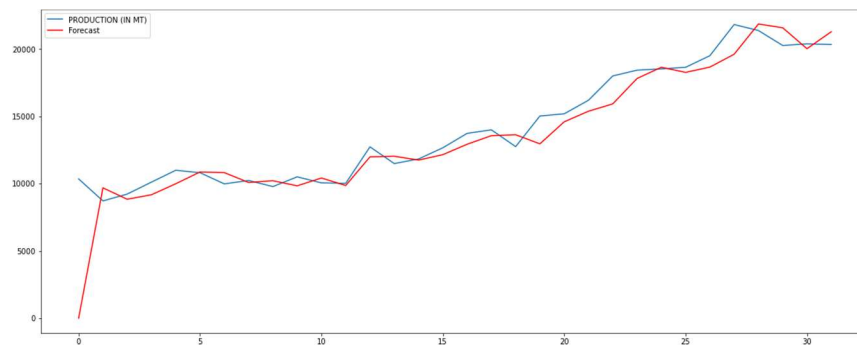


FIGURE 16. Yield Prediction showing Actual and Predicted Production

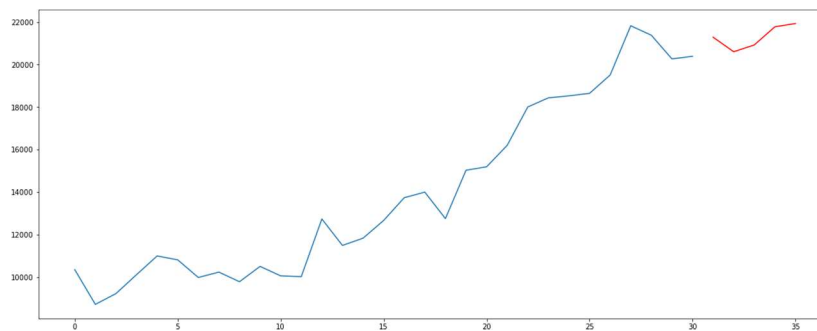


FIGURE 17. Forecasting yield

	year	predicted_mean
31	2022	21285.414550
32	2023	20600.349570
33	2024	20920.524292
34	2025	21774.952988
35	2026	21928.183103

FIGURE 18. Yield forecasting for the next 5 years

CONCLUSION

The project successfully works for yield estimation and for forecasting future yields with high speed and efficiency. Fruit yield estimation system is used to estimate the [11] accurate number of fruits present in the orchard. On video footage, fruit yield estimation is done by first identifying and then tracking the fruits in each frame of the input video. Bounding boxes are recovered from the YOLO V5 model, and Non-Maximum Suppression (NMS) is employed to induce final detections. The tracking pipeline is then updated with the results. DeepSORT algorithm is used for tracking the fruits. By keeping track of the fruits throughout the video frames [1], it is ensured that we are estimating the total fruit count appropriately when they are detected. Future fruit yield prediction is also carried out using time series forecasting by ARIMA model. This work may greatly help the needy farmers in yield estimation and predicting the future yields for sustainable growth.

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